Replication of "An RBC Model with Growth: The Role of Human Capital"

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Summary

This report replicates and analyzes the core results of Einarsson and Marquis (1998), who extend the standard Real Business Cycle framework by introducing endogenous human capital accumulation through time allocation between work, leisure, and training. We had two objective: to verify whether the model reproduces the qualitative empirical patterns emphasized in the paper, particularly regarding labor volatility and productivity-labor comovement; and to assess how the introduction of stochastic human capital (Model B) alters the cyclical properties of the economy relative to both a standard RBC model and the deterministic human-capital model (Model A). Using Dynare simulations of the stationary detrended system, we compute second moments and impulse responses and compare them to the paper's reported results and their U.S. data. Despite not reporting the standard error across our simulations (compared to the paper), our replication successfully captures the economic mechanisms highlighted by the authors.

Introduction

The goal of this paper was to replicate the results of Einarsson and Marquis 1998. Their contribution shows that understanding the interaction between human capital accumulation, labor supply, and long-run growth is central to macroeconomic theory. In their research, Einarsson and Marquis develop a Real business cycle in which human capital (or the value labor/ labor effiency) evolves endogenously through time allocation decisions, allowing them to study how productivity shocks spread through both physical and human capital. The authors show that endogenous human capital accumulation alters the dynamic response of the economy to productivity shocks. They focus on time allocation between work, leisure, and training, and how these specifications introduce additional spreading dynamics that affect output, investment, and labor relative to standard RBC models.

Model

The setting for the model is an economy composed of identical infinitely lived households, which derive their utility from consumption and leisure. They suppose that the representative household agent is either enjoying leisure time or contributing to production with nonleisure time. However, nonleisure time does not infer labor supply, as the agent has access to two types of production technologies, which require nonleisure time as input. The first one is working to produce output, which increases the household's stock of human capital via "learning by doing". The second one is to sacrifice production labor supply and follow formal training, which will increases human capital accumulation at a faster rate. The representative household maximizes his lifetime utility:

$$\max \ \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t, \ell_t), \qquad \beta \in (0,1)$$

Where β is the discount factor, and c_t, ℓ_t are consumption and leisure. The utility function is defined as the following logarithmic form:

$$u(c_t, \ell_t) = \ln(c_t) + \eta \ln(\ell_t), \qquad \eta > 0$$

Constraints

The maximisation problem is subject to three constraints (two ressource constraints and one human capital accumulation).

Consumption Constraint

The first one is the consumption constraint:

$$c_t = f(k_t, s_t n_t h_t, z_t) + (1-\delta)k_t - k_{t+1}, \qquad s_t \in (0,1)$$

where $f(.)$ is a production function, k_t and h_t are the physical and human capital stock at time t, n_t represents nonleisure time, z_t is a random productivity shock that follows an AR(1) Markov process, and s_t is the allocation of nonleisure time devoted to **production**.

Time Constraint

The second constraint is time:

$$n_t = \bar{n} - \ell_t, \qquad \ell_t \in [0,1]$$

where $\bar{n}$ is total time endowment.

Human Capital Constraint

The third constraint is the human capital production technology:

$$h_{t+1} = g(h_t, n_t, s_t), \qquad g_h, g_n > 0, \qquad g_s < 0, \qquad g_{hh} = 0$$

So the production technology is increasing in h_t and decreasing in s_t, as more time allocated to working. The $g_{hh} = 0$ assumption is the main element of endogeneous growth in the model.

Maximization Problem

The social planner faces the following maximization problem:

$$\max_{\{k_{t+1}, n_t, s_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \, u\left[\underbrace{f(k_t, s_t n_t h_t, z_t) + (1-\delta)k_t - k_{t+1}}_{=c_t}, \; \underbrace{\bar{n} - n_t}_{=\ell_t}\right] \qquad \text{given } h_0, k_0 > 0$$

subject to:

$$h_{t+1} = g(h_t, n_t, s_t)$$

where k_t, h_t are the state variables (choosing them today affects their value tomorrow), and c_t, ℓ_t, n_t, s_t are the control variables (choosed every period). It is also required that $c_t, \ell_t, h_{t+1}, k_{t+1} \geq 0$.

Bellman

The problem can be solved recursively with a Bellman function:

$$V(h_t, k_t, z_t) = \max_{\{k_{t+1}, n_t, s_t\}} \left\{ u(c_t, \ell_t) + \beta \int V(\underbrace{h_{t+1}}_{=g(h_t, n_t, s_t)}, k_{t+1}, z_{t+1}) \underbrace{Q(z_t, dz_{t+1})}_{\ln(z_t) = \theta \ln(z_{t+1}) + \varepsilon_t} \right\}.$$

which yields the following euler equations:

$$\textbf{Euler 1:} \qquad u_{c_t} = \beta \mathbb{E}_t[u_{c_{t+1}}(f_{k_{t+1}}(.) + 1 - \delta)]$$

$$\textbf{Euler 2:} \qquad u_{c_t} f_{s_t} + \beta g_{s_t} \mathbb{E}_t \left[u_{c_{t+1}} [f_{h_{t+1}} - \frac{f_{s_{t+1}}}{g_{s_{t+1}}} g_{h_{t+1}}] \right] = 0$$

$$\textbf{Euler 3:} \qquad u_{c_t} f_{n_t} - u_{\ell_t} + \beta g_{n_t} \mathbb{E}_t \left[u_{c_{t+1}} (f_{h_{t+1}} - g_{h_{t+1}} \frac{f_{n_{t+1}}}{g_{n_{t+1}}}) + u_{\ell_{t+1}} \frac{g_{h_{t+1}}}{g_{n_{t+1}}} \right] = 0$$

The derivations to obtain the result shown above can be found in the Bellman FOC and Euler derivations. **Euler 1** states that the household is indifferent between consuming one unit today and saving it as capital to consume tomorrow, **Euler 2** confronts the household with: "Should I use an additional unit of working time to produce output today, or stock human capital for tommorow?" and **Euler 3** lays the price (working more today, sacrificing leisure) for higher income and faster human capital accumulation tomorrow.

Functional forms

As decribed earlier, the utility has the following function:

$$u(c_t, \ell_t) = \ln c_t + \eta \ln \ell_t, \qquad \eta > 0$$

The production function $f(.)$ and A_t are defined as:

$$f(k_t, s_t n_t h_t, z_t) = z_t A_t k_t^{\alpha_1} (s_t h_t z_t)^{\alpha_2}, \qquad A_t = A_0 (1+\lambda)^t, \qquad \alpha_1, \alpha_2 \in (0,1) \qquad \text{and} \qquad \lambda \geq 0$$

and the production for human capital accumulation is:

$$h_{t+1} = g(h_t, n_t, s_t) = h_t \left[\rho_1 (1 - s_t) n_t + \rho_2 n_t s_t + 1 - \delta^h \right] \qquad \rho_1, \rho_2 > 0, \qquad \rho_1 > \rho_2, \qquad \delta^h \in (0,1)$$

The eulers written with their functional forms are:

Euler 1

$$\frac{1}{c_t} = \beta \mathbb{E}_t [\frac{1}{c_{t+1}} (z_{t+1} A_{t+1} \alpha_1 k_{t+1}^{\alpha_1 - 1} (h_{t+1} n_{t+1} s_{t+1})^{\alpha_2} + 1 - \delta)]$$

Euler 2

$$\frac{1}{c_t} Y_t n_t h_t + \beta \left[h_t n_t (\rho_2 - \rho_1) \right] \mathbb{E}_t \left[\frac{1}{c_{t+1}} Y_{t+1} h_{t+1} \left(s_{t+1} - \frac{\rho_1 (1 - s_{t+1}) n_{t+1} + \rho_2 s_{t+1} n_{t+1} + 1 - \delta^h}{n_{t+1} (\rho_2 - \rho_1)} \right) \right] = 0$$

Euler 3

$$\frac{1}{c_t} Y_t s_t n_t - \frac{\eta}{\ell_t} + \beta \left[h_t \rho_1 (1 - s_t) + \rho_2 s_t \right] \mathbb{E}_t \left[\frac{1}{c_{t+1}} \left[Y_{t+1} s_{t+1} \right] \left[n_{t+1} - h_{t+1} X + \frac{\eta}{\ell_{t+1}} \frac{X}{Y_{t+1}} \right] \right] = 0$$

$$\text{With: } X = \frac{\rho_1 (1 - s_{t+1}) n_{t+1} + \rho_2 s_{t+1} n_{t+1} + 1 - \delta^h}{n_{t+1} \rho_1 (1 - s_{t+1}) + \rho_2 s_{t+1}}, \qquad Y_{t+1} = \alpha_2 z_{t+1} A_{t+1} k_{t+1}^{\alpha_1} (s_{t+1} n_{t+1} h_{t+1})^{\alpha_2 - 1}$$

$$\text{and} \qquad Y_t = \alpha_2 z_t A_t k_t^{\alpha_1} (s_t n_t h_t)^{\alpha_2 - 1}$$

Please note that the notation used for X and Y_t, Y_{t+1} are only for simpler writing (these **notations will be redefined** in an upcoming subsection).

Model equations

Variables: $c, k, h, n, s, \ell, y, I, z$ Production function with $\alpha_1 + 2\alpha_2 = 1$.

(1) $y_t = z_t\, A_t\, k_t^{\alpha_1} \left(h_t n_t s_t\right)^{\alpha_2}$ Production function

(2) $n_t = \bar{n} - \ell_t$ Time constraint

(3) $i_t = k_{t+1} - (1-\delta)k_t$ Physical capital accumulation

(4) $y_t = c_t + i_t$ Resource constraint

(5) $h_{t+1} = h_t\Big[1 - \delta^h + \rho_1(1-s_t)n_t + \rho_2 n_t s_t\Big]$ $\rho_1, \rho_2 > 0$ Human capital accumulation

(6) $\ln(z_t) = \rho \ln(z_{t-1}) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ Productivity shock

(7) $u_{c_t} = \beta\, \mathbb{E}_t\big[u_{c_{t+1}}\big(1 - \delta + f_{k_{t+1}}\big)\big]$ Consumption Euler equation

(8) $u_{c_t} f_{s_t} + \beta\, g_{s_t}\, \mathbb{E}_t\left[u_{c_{t+1}}\left(f_{h_{t+1}} - f_{s_{t+1}} \dfrac{g_{h_{t+1}}}{g_{s_{t+1}}}\right)\right] = 0$ Training-work time share Euler (s_t)

(9) $u_{c_t} f_{n_t} - u_{\ell_t} + \beta\, g_{s_t}\, \mathbb{E}_t\left[u_{c_{t+1}}\left(f_{h_{t+1}} - f_{s_{t+1}} \dfrac{g_{h_{t+1}}}{g_{s_{t+1}}}\right) + u_{\ell_{t+1}} \dfrac{g_{h_{t+1}}}{g_{\ell_{t+1}}}\right] = 0$ Non leisure - leisure Euler (n_t)

Functional form derivatives

$$
\begin{aligned}
u(c_t, \ell_t) &= \ln(c_t) + \eta \ln(\ell_t) \\
u_c(c_t, \ell_t) &= \frac{1}{c_t}, \qquad u_\ell(c_t, \ell_t) = \frac{\eta}{\ell_t} \\
f(k_t, h_t n_t s_t, z_t) &= z_t A_t k_t^{\alpha_1} \left(h_t n_t s_t\right)^{\alpha_2}
\end{aligned}
$$

Production marginal products:

$$
\begin{aligned}
f_{k_t}(\cdot) &= \alpha_1 z_t A_t k_t^{\alpha_1 - 1} (h_t n_t s_t)^{\alpha_2} = \alpha_1 \frac{y_t}{k_t} \\
f_{s_t}(\cdot) &= \alpha_2 z_t A_t k_t^{\alpha_1} (h_t n_t s_t)^{\alpha_2 - 1}\, n_t h_t = \alpha_2 \frac{y_t}{s_t} \\
f_{n_t}(\cdot) &= \alpha_2 z_t A_t k_t^{\alpha_1} (h_t n_t s_t)^{\alpha_2 - 1}\, s_t h_t = \alpha_2 \frac{y_t}{n_t} \\
f_{h_t}(\cdot) &= \alpha_2 z_t A_t k_t^{\alpha_1} (h_t n_t s_t)^{\alpha_2 - 1}\, s_t n_t = \alpha_2 \frac{y_t}{h_t}
\end{aligned}
$$

Human-capital growth marginal products:

$$
\begin{aligned}
h_{t+1} &= g(h_t, n_t, s_t) = h_t\Big[\rho_1(1-s_t)n_t + \rho_2 n_t s_t + 1 - \delta^h\Big] \\
g_{s_t}(\cdot) &= \rho_2 h_t n_t - \rho_1 h_t n_t = (\rho_2 - \rho_1) h_t n_t \\
g_{n_t}(\cdot) &= h_t[\rho_1(1-s_t) + \rho_2 s_t] \\
g_{h_t}(\cdot) &= \rho_1(1-s_t)n_t + \rho_2 n_t s_t + 1 - \delta^h
\end{aligned}
$$

Log-Linearisation

We show in the mathematical appendix the detailed Log linearisation for 3 equations.

Stationarity issue

We first had to deal with A_t:

$$A_t = A_0(1+\lambda)^t \rightarrow \text{exogenous deterministic technology trend} \rightarrow \text{must handle stationarity.}$$

A_t enters multiplicatively in the production function, so a rise in A_t scales up output and the marginal products over time. As a result, the optimal paths of output y_t , consumption c_t and capital k_t do not converge to constant levels: they grow without bound → there is no constant Steady State. We therefore rewrite the model in stationary form by detrending these variables with an appropriate growth factor X_t, so the transformed "hat" variables are constant along the balanced growth path and Dynare can compute the steady state. e.g.

$$\hat{y}_t = \frac{y_t}{X_t}$$

Proposed trend:

$$X_t = A_t^{\frac{1}{1-\alpha_1}} \, h_t^{\frac{\alpha_2}{1-\alpha_1}}$$

Detrending algebra:

$$\begin{aligned}
y_t &= z_t A_t k_t^{\alpha_1} (n_t s_t)^{\alpha_2} h_t^{\alpha_2} \\
\hat{k}_t = \frac{k_t}{X_t} &\iff k_t = \hat{k}_t X_t \quad \text{and} \quad \hat{y}_t = \frac{y_t}{X_t} \\
\hat{y}_t X_t = z_t A_t (\hat{k}_t X_t)^{\alpha_1} (s_t n_t)^{\alpha_2} h_t^{\alpha_2} &\Leftrightarrow \hat{y}_t = z_t \hat{k}_t^{\alpha_1} (s_t n_t)^{\alpha_2} A_t X_t^{\alpha_1 - 1} h_t^{\alpha_2}
\end{aligned}$$

Condition:

$$A_t h_t^{\alpha_2} X_t^{\alpha_1 - 1} = 1 \quad \iff \quad X_t = \left(A_t h_t^{\alpha_2}\right)^{\frac{1}{1-\alpha_1}} = A_t^{\frac{1}{1-\alpha_1}} \, h_t^{\frac{\alpha_2}{1-\alpha_1}}$$

To have:

$$\hat{y}_t = z_t \hat{k}_t^{\alpha_1} (s_t n_t)^{\alpha_2}$$

Detrended system

Using the above definition for X_t and the detrended form $\tilde{x}_t = \frac{x_t}{X_t}$ where $x_t = (y_t, k_t, c_t, i_t)$, we can rewrite the system of equations in the detrended form:

Production function (1)

$$\hat{y}_t = z_t \, \hat{k}_t^{\alpha_1} \, (s_t n_t)^{\alpha_2}$$

Resource constraints (2)

$$\begin{aligned}
\hat{y}_t = \frac{y_t}{X_t}, \hat{c}_t = \frac{c_t}{X_t}, \; \hat{i}_t = \frac{i_t}{X_t} \\
\Rightarrow \hat{y}_t = \hat{c}_t + \hat{i}_t
\end{aligned}$$

Time constraint (3)

$$n_t = \bar{n} - \ell_t.$$

Capital accumulation (4)

$$\begin{aligned}
k_{t+1} &= (1-\delta) k_t + i_t \\
\hat{k}_{t+1} X_{t+1} = (1-\delta)\hat{k}_t X_t + \hat{i}_t X_t &= \frac{(1-\delta)\hat{k}_t X_t + \hat{i}_t X_t}{X_{t+1}} = \frac{(1-\delta)\hat{k}_t + \hat{i}_t}{X_{t+1}/X_t}.
\end{aligned}$$

We denote:

$$g_t^x = \frac{X_{t+1}}{X_t} \qquad \Rightarrow \qquad \hat{k}_{t+1} = \frac{(1-\delta)\hat{k}_t + \hat{i}_t}{g_t^x}$$

Human capital accumulation (5)

$$\begin{aligned}
h_{t+1} &= \Big[1 - \delta^h + \rho_1 (1 - s_t) n_t + \rho_2 s_t n_t\Big] h_t \\
\Rightarrow G_t = \frac{h_{t+1}}{h_t} &= 1 - \delta_h + \rho_1 (1 - s_t) n_t + \rho_2 s_t n_t
\end{aligned}$$

Shock process (6)

$$\ln(z_t) = \rho \ln(z_{t-1}) + \varepsilon_t$$

Euler equations

Euler 1 (7)

$$\frac{1}{c_t} = \beta\,\mathbb{E}_t\left[\frac{1}{c_{t+1}}\left(1-\delta+f_{k_{t+1}}\right)\right]$$

With

$$f_{k_{t+1}} = \alpha_1 z_{t+1} A_{t+1} k_{t+1}^{\alpha_1-1}\left(s_{t+1}n_{t+1}h_{t+1}\right)^{\alpha_2} = \alpha_1\frac{y_{t+1}}{k_{t+1}},$$

by multiplying Euler 1 by X_t and replacing by the variable in "hat" form, the detrended form can written as:

$$\frac{1}{\hat{c}_t} = \beta\,\mathbb{E}_t\left[\frac{1}{\hat{c}_{t+1}\,g_t^x}\left(1-\delta+\alpha_1\frac{\hat{y}_{t+1}}{\hat{k}_{t+1}}\right)\right].$$

Euler 2 (8)

As written in derivative form:

$$-u_{c_t}f_{s_t} = \beta\,g_{s_t}\,\mathbb{E}_t\left[u_{c_{t+1}}\left(f_{h_{t+1}} - f_{s_{t+1}}\frac{g_{h_{t+1}}}{g_{s_{t+1}}}\right)\right].$$

Substituting the hat form variables and plugging in:

$$u_{c_t} = \frac{1}{c_t}, \qquad f_{s_t} = \alpha_2\frac{y_t}{s_t}, \qquad f_{h_{t+1}} = \alpha_2\frac{y_{t+1}}{h_{t+1}}, \qquad g_{s_t} = h_t n_t(\rho_2-\rho_1), \qquad \frac{g_{h_{t+1}}}{g_{s_{t+1}}} = \frac{G_{t+1}}{h_{t+1}n_{t+1}(\rho_2-\rho_1)}$$

yields:

$$-\frac{1}{c_t}\,\alpha_2\frac{y_t}{s_t} = \beta\,h_t n_t(\rho_2-\rho_1)\;\mathbb{E}_t\left[\frac{1}{c_{t+1}}\left(\alpha_2\frac{y_{t+1}}{h_{t+1}} - \alpha_2\frac{y_{t+1}}{s_{t+1}}\frac{G_{t+1}}{h_{t+1}n_{t+1}(\rho_2-\rho_1)}\right)\right]$$

which can be rearranged as:

$$-\frac{\hat{y}_t}{\hat{c}_t}\,\alpha_2\frac{1}{s_t} = \beta\,n_t(\rho_2-\rho_1)\,\mathbb{E}_t\left[\alpha_2\frac{\hat{y}_{t+1}}{\hat{c}_{t+1}G_t}\left(1-\frac{G_{t+1}}{s_{t+1}n_{t+1}(\rho_2-\rho_1)}\right)\right]$$

Euler 3 (9)

Derivative form:

$$u_{c_t}f_{n_t} + \beta g_{s_t}\mathbb{E}_t\left[u_{c_{t+1}}\left(f_{h_{t+1}} - f_{n_{t+1}}\frac{g_{h_{t+1}}}{g_{n_{t+1}}}\right) + u_{\ell_{t+1}}\frac{g_{h_{t+1}}}{g_{n_{t+1}}}\right] = u_{\ell_t}.$$

substituting $u_c = 1/c$, $u_\ell = \eta/\ell$, the hat form and using the same $f_h, f_s, g_h/g_s$ objects as above, produces:

$$\frac{\hat{y}_t}{\hat{c}_t}\frac{\alpha_2}{n_t} + \beta\big[\rho_1(1-s_t)+\rho_2 s_t\big]\,\mathbb{E}_t\frac{1}{G_t}\left[\frac{\hat{y}_{t+1}}{\hat{c}_{t+1}}\alpha_2\Big(1-\frac{G_{t+1}}{n_{t+1}[\rho_1(1-s_{t+1})+\rho_2 s_{t+1}]}\Big) + \frac{\eta}{\ell_{t+1}}\frac{G_{t+1}}{\rho_1(1-s_{t+1})+\rho_2 s_{t+1}}\right] = \frac{\eta}{\ell_t},$$

Trend growth factor (10)

$$\text{with} \quad \Gamma = 1+\lambda \qquad \text{and} \qquad G_t = \frac{h_{t+1}}{h_t}.$$

$$\Rightarrow \quad g_t^x = \Gamma^{\frac{1}{1-\alpha_1}}\,G_t^{\frac{\alpha_2}{1-\alpha_1}}.$$

Steady state

In this model, we could solve several steady-state relationships analytically. However, the optimality conditions of Euler 2 for n and Euler 3 for s can only be solve numerically. We therefore provide initial guesses for n and s in `initval`, and Dynare's `steady` routine solves the full nonlinear steady-state system numerically, adjusting n and s until all equilibrium conditions hold. (See Appendix B - Dynare code)

- $Y = C + I \qquad Y = ZAK^{\alpha_1}(SNH)^{\alpha_2}$
- $N = \bar{N} - L$
- $lnZ = \rho lnZ \Leftrightarrow lnZ(1-\rho) = 0 \Leftrightarrow lnZ = 0 \Leftrightarrow Z = 1$
- $I = K - (1-\delta)K = K\delta$
- $G = \left[1 - \delta^h + p_1(1-S)N + p_2 SN\right]$
- $g^x = \Gamma^{\frac{1}{1-\alpha_1}} G^{\frac{\alpha_2}{1-\alpha_1}}$
- $K = \frac{(1-\delta)K+I}{g^x} \;\Leftrightarrow\; g^x K = (1-\delta)K + I \;\Leftrightarrow\; K\left(g^x - 1 + \delta\right) = I.$

Euler 1 (steady state)

$$\begin{aligned} & \frac{1}{C} = \beta \frac{1}{C} \frac{1}{g_x} \left(1 - \delta + \alpha_1 \frac{Y}{K}\right), \\ \Leftrightarrow \quad & g_x = \beta \left(1 - \delta + \alpha_1 \frac{Y}{K}\right), \\ \Leftrightarrow \quad & \frac{Y}{K} = \frac{\frac{g_x}{\beta} - 1 + \delta}{\alpha_1}. \end{aligned}$$

Euler 2 (steady state)

$$\begin{aligned} & -\frac{Y}{C}\frac{1}{S}\alpha_2 = \beta N(p_2 - p_1)\frac{Y}{C}\frac{1}{G}\alpha_2 \left[1 - \frac{G}{SN(p_2 - p_1)}\right], \\ \Leftrightarrow \quad & -\frac{1}{S} = \beta N(p_2 - p_1)\frac{1}{G} - \beta\frac{1}{S}, \\ \Leftrightarrow \quad & (p_1 - p_2)\,\beta N \frac{1}{G} = \frac{1}{S}(1-\beta), \\ \Leftrightarrow \quad & S = \frac{(1-\beta)G}{(p_1 - p_2)\beta N}. \end{aligned}$$

Euler 3 (steady state)

$$\begin{aligned} & \frac{1}{C}\frac{Y}{N}\alpha_2 + \beta\big(p_1(1-S) + p_2 S\big)\frac{1}{G}\left\{\frac{Y}{C}\alpha_2\left[1 - \frac{G}{N\big(p_1(1-S) + p_2 S\big)}\right] + \frac{\eta}{L}\frac{G}{\big(p_1(1-s) + p_2 s\big)}\right\} = \frac{\eta}{L}, \\ \Leftrightarrow \quad & \frac{1}{C}\frac{Y}{N}\alpha_2 + \beta\left\{\frac{Y}{C}\alpha_2\left[\frac{p_1(1-S) + p_2 S}{G} - \frac{1}{N}\right] + \frac{\eta}{L}\right\} = \frac{\eta}{L}. \end{aligned}$$

Our results versus the paper's

You can find a copy of our code in Appendix B - Dynare code.

The problem we had

The paper specifies the structure that we described above, but does not report an explicit numerical value for $\bar{n}$, so $\bar{n}$ acts as a normalization of time units that must be chosen to keep the steady state economically admissible under the paper's calibration. With the reported annual parameters for human-capital accumulation and preferences, setting $\bar{n}$ too low (close to 1) can push the unconstrained solution outside the feasible region (e.g. $s > 1$, implying negative training time and implausible human-capital dynamics). We therefore set $\bar{n} = 2$, which yields an interior steady state with $s \approx 0.88 \in (0, 1)$ and positive human-capital growth ($G > 1$), preserving the intended interpretation of time allocation while allowing comparisons of simulated second moments with the paper's results.

IRFs

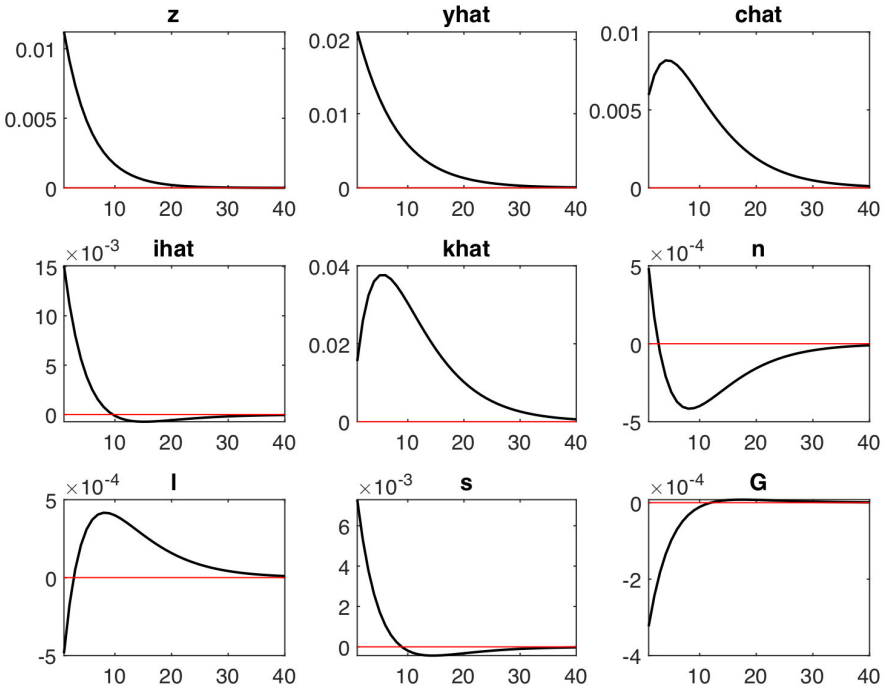


Figure 1: Model A IRFs for z, $\hat{y}, \hat{c}, \hat{i}, \hat{k}, n, \ell, s, G$

- z_t jumps after an unexpected increase in productivity, it is the source of all other responses.
- $\hat{y}$ (production output): Goes up immediately because a higher productivity level raises production for a given capital and labor. It then declines slowly back to the steady state with persistence dictated by capital adjustment. Capital changes slowly so output remains elevated even after the initial shock dies out.
- $\hat{c}$ (consumption): It is chosen via the Euler equation 1, it increases on impact because households are temporarily richer as output and income rise. The hump shape reveals that there is also a share that is saved for consumption smoothing. Consumers dislike large jumps in consumption, so they will invest to share these gains overtime.
- $\hat{i}$ (investment): It is the part of the output that is devoted to accumulating capital. Firstly, we see that investment is more volatile than output. Secondly, the productivity shock increases the marginal product of capital, investing today yields a higher return tomorrow so it increases on impact. It then decreases as the productivity shock dies out and as accumulated capital reduces return via diminishing marginal productivity.

- $\hat{k}$ (physical capital): Typical hump shape shows that it goes up with the shock, and rises even more later as investment is high in the first periods (i.e. capital accumulates via its law of motion), then progressively returns back to its steady state value.

- n (nonleisure time): As it can be either work/learning by doing or training, this IRF shows the tension between the substitution effect SE and the wealth effect WE. When productivity is high, working is more valuable, so in the beginning people want to work more, pushing n up initially (SE). However, higher productivity also makes household richer in total wealth, so they want more leisure (WE), which explains the steep decrease in n.

- ℓ (leisure): Decreases on impact as people want to work more suddenly with the productivity shock, and then increases because of the wealth effect (has the inverse IRF of n) and slowly reverts back to SS.

- s (share of non leisure time allocated to production): With the productivity shock, workers want to dedicate more of their time to production and s goes up. However households now face an intertemporal tradeoff, as increasing s reduces formal training which slows human capital accumulation. Then, households gradually rebalance towards training and s declines back to its steady state value.

- G (gross human capital growth factor): Decreases immediately as workers devote more nonleisure time to production, and since $\rho_1 > \rho_2$ (learning by doing increases h_{t+1} less than formal training), the effective human capital accumulation is reduced. It then goes back up as workers want to accumulate more human capital.

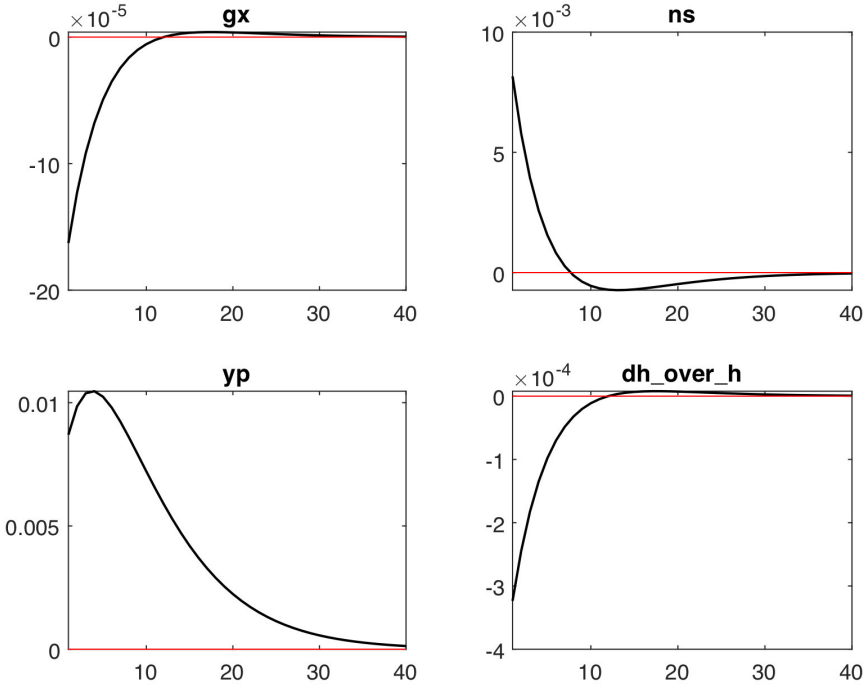


Figure 2: Model A IRFs for$g^x, ns, Y_p, \frac{\Delta h}{h}$

- $\frac{\Delta h}{h}$: Evolves the same way as G.

- g^x(growth rate of the detrending factor): depends positively on G, it evolves similarly. The economy's long term growth component tied to human capital temporarily weakens because the household changes their allocation of nonleisure time.

- ns (effective hours in production): Rises strongly on impact as household adjust not only the amount of time spent working but also reduces the share spent on formal training. It then decreases with

the alternating choices of nonleisure time, and rises again once workers are trained and supply labor production later in time.

- Y_p (productivity/production per capita): A big increase at first due to the productivity shock because each unit of labor produces more output. However it increases less than output because ns rises at the same time and dampens the increase of Y_p. As z_t reverts and labor normalizes, it gradually converges back to its steady state.

In our code, here are the modifications for Model B (please follow the instructions in the code for which lines to uncomment in order to run for Model B):

$$\textbf{Stochastic human-capital shock:} \quad \log(\gamma_{h_t}) = \rho_h \log(\gamma_{h_{t-1}}) + \varepsilon_{h_t},$$

$$\textbf{Human-capital gross growth (Model B):} \quad G_t = 1 - \gamma_{h_t}\,\delta_h + \big(\rho_1(1-s_t) + \rho_2 s_t\big) n_t.$$

Our Table 1 shows our results for our simulation of this model and the model with stochastic human capital:

Table 1: Second moments from Dynare simulations - Model A vs Model B

	Model A		Model B	
Variable	σ_x	$\rho_{x\hat{y}}$	σ_x	$\rho_{x\hat{y}}$
z	0.0154	0.9880	0.0166	0.9867
$\hat{y}$	0.0305	1.0000	0.0352	1.0000
$\hat{c}$	0.0138	0.8530	0.0192	0.9042
$\hat{i}$	0.0201	0.9338	0.0197	0.9094
$\hat{k}$	0.0600	0.6887	0.0839	0.8106
n	0.0009	0.3381	0.0013	-0.2857
l	0.0009	-0.3381	0.0013	0.2857
s	0.0097	0.9251	0.0099	0.9049
G	0.0004	-0.9531	0.0009	-0.0329
g^x	0.0002	-0.9531	0.0005	-0.0329
ns	0.0109	0.9021	0.0106	0.8655
yp	0.0179	0.8952	0.0241	0.9267
$\frac{\Delta h}{h}$	0.0004	-0.9531	0.0009	-0.0329
γ_h			0.0129	-0.3970
$\rho_{ns,yp}$	0.6153		0.6138	

Results Interpretation

The standard RBC model reported in Einarsson and Marquis exhibits very low volatility in hours worked, a well-known shortcoming of the framework. In our replication, the introduction of endogenous human capital accumulation increases labor volatility relative to the RBC benchmark. In Model A, the standard deviation of labor services is $\sigma_{ns} = 0.0109$, compared to a lower volatility in the standard RBC model reported in the paper (0.0052). However, our value is below both the paper's Model A result ($\sigma_{ns} = 0.0164$) and their U.S. data (0.0160), so labor fluctuations are underestimated. Regarding productivity–output comovement, the standard RBC model produces an excessively high correlation, close to unity. Our Model A reduces this correlation, with $\rho_{yp,\hat{y}} = 0.8952$, improving upon the RBC benchmark (0.9915) and moving closer to their US data result (0.792). Still, this correlation remains higher than in the paper's Model A simulation (0.8102), suggesting that productivity remains tightly linked to output in our version. A similar pattern emerges for the correlation between productivity and hours worked. In the RBC model, this correlation is very high (0.7915). Our Model A lowers this correlation to $\rho_{ns,yp} = 0.6153$. However, this value still exceeds the paper's Model A result (0.5120), indicating that our take on labor productivity remains overly procyclical. Finally, introducing stochastic human capital in Model B alters the cyclical behavior of human capital investment. In our simulation, the correlation between net human capital growth and output falls drastically from -0.9531 in Model A to -0.0329 in Model B, a much stronger reduction than in the paper, where the corresponding correlation decreases from -0.9106 to -0.4268. With our number being even closer to their US data simulation (-0.187)

than the paper's. This confirms that exogenous shocks on human capital accumulation weakens the countercyclicality implied by the deterministic version (Model A).

Discussion and Conclusion

Discussion

This replication exercise follows a simplified solution strategy relative to the original paper. As we rely on a steady-state computation, we only log-linearized three equations as this was a requirement. We encountered trouble with the value of $\bar{n}$, which was previously mentioned at the beginning of our results section. Moreover, for feasibility we imposed a stationarization (detrending) scheme to compute a stationary steady state, which may differ from the approach used in the original model; this mismatch is a plausible reason why we needed to set $\bar{n} = 2$ to obtain a well-behaved solution. It is also important to mention that our simulation results are based on multiple simulation paths. We report point estimates of averaged simulated second moments but we do not compute standard deviations **across simulations**, which the original paper does. We were not sure how to obtain this element of the paper, and as they were small in their results, so we decided to not include them. We believe that the absence of these standard errors does not prevent meaningful comparison, but hides the robustness of our results. Despite this main limitation, our results are informative. The qualitative mechanisms emphasized in the paper, namely the role of human capital accumulation and stochastic depreciation in shaping labor dynamics and productivity, are reproduced. The comparison between the standard RBC benchmark, Model A, and Model B shows that introducing human capital channels alters the volatility and comovement of labor input and productivity. Therefore, while our results should be interpreted with caution, our replication successfully confirms the economic insights of the original study.

Conclusion

Our aim was to replicate and understand the main mechanisms of Einarsson and Marquis by extending a standard RBC to include human capital accumulation and stochastic shocks. Using Dynare simulations and a first-order approximation around the steady state, we compared their benchmark RBC model with our results of their two augmented specifications. Our results confirm their initial insight: introducing human capital dynamics improves the model's ability to account for labor market fluctuations and productivity comovements relative to the standard RBC. Both Model A and Model B reduce the tight link between productivity and hours worked, while the inclusion of the stochastic human capital shock in Model B further weakens the correlation between human capital investment and output. Although our results differ from those reported in the paper (partly due to simplified simulation procedures), the qualitative patterns are there. To conclude, this replication highlights the importance of human capital as a state variable in business cycle analysis and illustrates how such extensions to the RBC framework can generate richer dynamics.

Appendix A - Mathematical derivations

Bellman FOC and Euler derivations

From the bellman maximimisation problem, we derive the following first order conditions:

1. w.r.t. k_{t+1}:
$$u_{c_t} = \beta \, \mathbb{E}_t \left[V_k(h_{t+1}, k_{t+1}, z_{t+1}) \right]$$
2. w.r.t. s_t:
$$u_{c_t} f_{s_t} = \beta g_{s_t} \, \mathbb{E}_t \left[V_h(h_{t+1}, k_{t+1}, z_{t+1}) \right]$$
3. w.r.t. n_t:
$$u_{c_t} f_{n_t} - u_{\ell_t} + \beta V_h(h_{t+1}, k_{t+1}, z_{t+1}) g_{n_t} = 0$$

Envelope Theorem

$$\text{The E.T. says that:} \quad \frac{\partial k^{\star}_{t+1}}{\partial k_t} = 0$$

For k_t:

$$V_k(h_t, k_t, z_t) = u_{c_t}\left(1 - \delta + f_{k_t} - \frac{\partial k^\star_{t+1}}{\partial k_t}\right) + \beta \frac{\partial V_{k_{t+1}}}{\partial k^\star_{t+1}} \frac{\partial k^\star_{t+1}}{\partial k_t}$$

$$\xrightarrow{\text{At t+1}} \quad V_k(h_{t+1}, k_{t+1}, z_{t+1}) = u_{c_{t+1}}\big(1 - \delta + f_{k_{t+1}}\big) \tag{4}$$

For h_t:

$$V_h(h_t, k_t, z_t) = u_{c_t} f_{h_t} + \beta \mathbb{E}_t \left[V_h(h_{t+1}, k_{t+1}, z_{t+1})\, g_h(h_t, n_t, s_t)\right]$$

$$\xrightarrow{\text{At t+1}} V_h(h_{t+1}, k_{t+1}, z_{t+1}) = u_{c_{t+1}} f_{h_{t+1}} + \beta \mathbb{E}_{t+1} \left[V_h(h_{t+2}, k_{t+2}, z_{t+2})\, g_h(h_{t+1}, n_{t+1}, s_{t+1})\right] \tag{5}$$

FOC for s_t to eliminate $\mathbb{E}_t[V_h]$

$$u_{c_t} f_{s_t} + \mathbb{E}_t \left[V_h(h_{t+1}, k_{t+1}, z_{t+1})\right] g_{s_t} = 0$$

$$\Rightarrow \quad \mathbb{E}_t[V_h(h_{t+1}, k_{t+1}, z_{t+1})] = -\frac{u_{c_{t+1}} f_{s_{t+1}}}{g_{s_{t+1}}} \tag{6}$$

$$V_h(h_{t+1}, k_{t+1}, z_{t+1}) = u_{c_{t+1}} f_{h_{t+1}} + \beta \mathbb{E}_{t+1} \left[V_h(h_{t+2}, k_{t+2}, z_{t+2})\right] g_h(h_{t+1}, n_{t+1}, s_{t+1})$$

Insert (6) into (5):

$$V_h(h_{t+1}, k_{t+1}, z_{t+1}) = u_{c_{t+1}} f_{h_{t+1}} + g_{h_{t+1}} \left(-\frac{u_{c_{t+1}} f_{s_{t+1}}}{g_{s_{t+1}}}\right) \tag{7a}$$

FOC for n_t to eliminate $\mathbb{E}_t[V_h]$

$$u_{c_t} f_{n_t} - u_{\ell_t} = \beta \mathbb{E}_t \left[V_h(h_{t+1}, k_{t+1}, z_{t+1})\right] g_{n_t} = 0$$

$$\Rightarrow \quad \beta \mathbb{E}_t \left[V_h(h_{t+1}, k_{t+1}, z_{t+1})\right] = \frac{u_{\ell_{t+1}} - u_{c_{t+1}} f_{n_{t+1}}}{g_{n_{t+1}}} \tag{8}$$

Combine (5) and (8):

$$V_h(h_{t+1}, k_{t+1}, z_{t+1}) = u_{c_{t+1}} f_{h_{t+1}} + g_{h_{t+1}} \left(\frac{u_{\ell_{t+1}} - u_{c_{t+1}} f_{n_{t+1}}}{g_{n_{t+1}}}\right) \tag{7b}$$

Euler Equations

$$(3) + (4) \Rightarrow \textbf{Euler 1:} \qquad u_{c_t} = \beta \mathbb{E}_t [u_{c_{t+1}} (f_{k_{t+1}}(.) + 1 - \delta)]$$

$$(2) + (7a) \Rightarrow \textbf{Euler 2:} \qquad u_{c_t} f_{s_t} + \beta g_{s_t} \mathbb{E}_t \left[u_{c_{t+1}} [f_{h_{t+1}} - \frac{f_{s_{t+1}}}{g_{s_{t+1}}} g_{h_{t+1}}]\right] = 0$$

$$(3) + (7b) \Rightarrow \textbf{Euler 3:} \qquad u_{c_t} f_{n_t} - u_{\ell_t} + \beta g_{n_t} \mathbb{E}_t \left[u_{c_{t+1}} (f_{h_{t+1}} - g_{h_{t+1}} \frac{f_{n_{t+1}}}{g_{n_{t+1}}}) + u_{\ell_{t+1}} \frac{g_{h_{t+1}}}{g_{n_{t+1}}}\right] = 0$$

Log linearisation

We use Taylor approximation with implicit functions and certainty equivalence (for Euler 1) to loglinearise.

Resource constraint

$$\Rightarrow \qquad Y_t = C_t + I_t, \qquad \text{SS} \Rightarrow \quad Y = C + I.$$

$$\Leftrightarrow \qquad \left(\frac{\partial Y}{\partial Y}\right)\left(\frac{Y_t - Y}{Y}\right)Y = \left(\frac{\partial C}{\partial C}\right)\left(\frac{C_t - C}{C}\right)C + \left(\frac{\partial I}{\partial I}\right)\left(\frac{I_t - I}{I}\right)I,$$

$$\Leftrightarrow \qquad Y\,\tilde{y}_t = C\,\tilde{c}_t + I\,\tilde{i}_t,$$

$\Downarrow$ divide by the steady state (SS)

$$\Leftrightarrow \qquad \tilde{y}_t = \frac{C}{Y}\,\tilde{c}_t + \frac{I}{Y}\,\tilde{i}_t.$$

Production function

$$\Rightarrow \quad y_t = z_t\,A_t\,k_t^{\alpha_1}\left(s_t n_t h_t\right)^{\alpha_2}, \qquad \text{SS} \Rightarrow \quad Y = Z\,A\,K^{\alpha_1}(SNH)^{\alpha_2}.$$

$$G(y_t) = H(z_t, A_t, k_t, s_t, n_t, h_t).$$

$$\begin{aligned} Y\,\tilde{y}_t = {} & AK^{\alpha_1}(SNH)^{\alpha_2}\,Z\,\tilde{z}_t \;+\; ZK^{\alpha_1}(SNH)^{\alpha_2}\,A\,\tilde{a}_t \;+\; \alpha_1 ZAK^{\alpha_1-1}(SNH)^{\alpha_2}\,K\,\tilde{k}_t \\ & +\; \alpha_2 ZAK^{\alpha_1}(SNH)^{\alpha_2-1}\,NH\,S\,\tilde{s}_t \;+\; \alpha_2 ZAK^{\alpha_1}(SNH)^{\alpha_2-1}\,SH\,N\,\tilde{n}_t \;+\; \alpha_2 ZAK^{\alpha_1}(SNH)^{\alpha_2-1}\,SN\,H\,\tilde{h}_t. \end{aligned}$$

$\Downarrow$ divide by SS

$$\begin{aligned} \tilde{y}_t &= \tilde{z}_t + \tilde{a}_t + \alpha_1\tilde{k}_t + \alpha_2\tilde{s}_t + \alpha_2\tilde{n}_t + \alpha_2\tilde{h}_t, \\ \tilde{y}_t &= \tilde{z}_t + \tilde{a}_t + \alpha_1\tilde{k}_t + \alpha_2\left(\tilde{s}_t + \tilde{n}_t + \tilde{h}_t\right). \end{aligned}$$

Euler 1

$$\Rightarrow \quad \frac{1}{\beta}\frac{\mathbb{E}_t C_{t+1}}{C_t} = 1 - \delta + \mathbb{E}_t\left[\alpha_1 z_{t+1} A_{t+1} k_{t+1}^{\alpha_1-1}(s_{t+1}n_{t+1}h_{t+1})^{\alpha_2}\right]$$

$$\text{SS} \Rightarrow \quad \frac{1}{\beta} + \delta - 1 = \alpha_1 ZAK^{\alpha_1-1}(SNH)^{\alpha_2}$$

$$G(C_{t+1}, C_t) = H(z_{t+1}, A_{t+1}, k_{t+1}, s_{t+1}, n_{t+1}, h_{t+1}).$$

$$\begin{aligned} \Leftrightarrow \quad \frac{C}{C}\,\tilde{c}_{t+1}\,\frac{1}{\beta} - \frac{1}{\beta}\,\frac{C}{C^2}\,C\,\tilde{c}_t = {} & \alpha_1 AK^{\alpha_1-1}(SNH)^{\alpha_2}\,Z\,\tilde{z}_{t+1} + \alpha_1 ZK^{\alpha_1-1}(SNH)^{\alpha_2}\,A\,\tilde{a}_{t+1} \\ & + \alpha_1 ZA(\alpha_1-1)K^{\alpha_1-2}(SNH)^{\alpha_2}\,K\,\tilde{k}_{t+1} + \alpha_1 ZAK^{\alpha_1-1}\alpha_2(SNH)^{\alpha_2-1}\,NH\,S\,\tilde{s}_{t+1} \\ & + \alpha_1 ZAK^{\alpha_1-1}\alpha_2(SNH)^{\alpha_2-1}\,SH\,N\,\tilde{n}_{t+1} + \alpha_1 ZAK^{\alpha_1-1}\alpha_2(SNH)^{\alpha_2-1}\,SN\,H\,\tilde{h}_{t+1} \end{aligned}$$

$$\Leftrightarrow \quad \frac{1}{\beta}\left(\tilde{c}_{t+1}-\tilde{c}_t\right) = \alpha_1 AK^{\alpha_1-1}Z(SNH)^{\alpha_2}\,\tilde{z}_{t+1} + \alpha_1 ZAK^{\alpha_1-1}(SNH)^{\alpha_2}\,\tilde{a}_{t+1}$$
$$+\alpha_1 ZAK^{\alpha_1-1}(SNH)^{\alpha_2}\,\tilde{k}_{t+1}(\alpha_1-1) + \alpha_1 ZAK^{\alpha_1-1}(SNH)^{\alpha_2}\,\alpha_2\,\tilde{s}_{t+1}$$
$$+\alpha_1 ZAK^{\alpha_1-1}(SNH)^{\alpha_2}\,\alpha_2\,\tilde{n}_{t+1} + \alpha_1 ZAK^{\alpha_1-1}(SNH)^{\alpha_2}\,\alpha_2\,\tilde{h}_{t+1}$$

$$\Downarrow \text{ divide by SS} = \alpha_1 ZAK^{\alpha_1-1}(SNH)^{\alpha_2}$$

$$\Leftrightarrow \quad \frac{1}{\beta}\left(\tilde{c}_{t+1}-\tilde{c}_t\right)\frac{1}{\alpha_1 ZAK^{\alpha_1-1}(SNH)^{\alpha_2}} = \tilde{z}_{t+1} + \tilde{a}_{t+1} + (\alpha_1-1)\tilde{k}_{t+1} + \alpha_2(\tilde{s}_{t+1}+\tilde{n}_{t+1}+\tilde{h}_{t+1})$$

$$\Leftrightarrow \quad \frac{1}{\beta}\left(\mathbb{E}_t\tilde{c}_{t+1}-\tilde{c}_t\right)\frac{1}{\frac{1}{\beta}+\delta-1} = \mathbb{E}_t\left[\tilde{z}_{t+1} + \tilde{a}_{t+1} + (\alpha_1-1)\tilde{k}_{t+1} + \alpha_2(\tilde{s}_{t+1}+\tilde{n}_{t+1}+\tilde{h}_{t+1})\right]$$

Appendix B - Dynare code

Listing 1: Dynare model: RBC + Human Capital (stationary hat model)

```
1
2  //============================================================
3  // RBC + Human Capital (stationary hat model)
4  // with paper-style observables: ns, y/ns, Dh/h
5  //============================================================
6
7  //------------------------------------------------------------
8  // 1) Declare variables and shocks
9  //------------------------------------------------------------
10 var z yhat chat ihat khat n l s G gx ns yp dh_over_h ;
11 varexo eps;
12 %add gammah in var and epsh in varexo to replicate model B%
13
14 //------------------------------------------------------------
15 // 2) Parameters
16 //------------------------------------------------------------
17 parameters alpha1 alpha2 beta eta delta deltah p1 p2 nbar rho sigma Gamma lambda;
18 %add rhoh and sigmah in parameters to replicate model B%
19
20 // Parameterisation (paper's "Model A & B" style annual values)
21
22 alpha1 = 0.3 ;             // capital share in production
23 alpha2 = 0.35 ;            // elasticity of output w.r.t. effective labor/training input
24 beta = 0.96;               // discount factor
25 delta = 0.060;             // physical capital depreciation rate
26 deltah = 0.050;            // human capital depreciation rate
27 p1 = 0.090;                // productivity of non-training time in H K accumulation
28 p2 = 0.045;                // productivity of training time in H K accumulation
29 eta = 0.87;                // weight of leisure in utility
30 rho = 0.81;                // persistence of AR1 TFP shock process
31 lambda = 0.0054;            // deterministic growth rate of exogenous technology At
32 Gamma = 1 + lambda;        // gross growth factor of technology.
33 sigma = 0.0112;            // std dev TFP innovation epsilon for model A %(to comment out for model B)%
34 nbar = 2;                  // total time endowment
35
36 %Model B parameters to uncomment%
37 //rhoh = rho;              //persistence of AR1 human capital shock process
38 //sigmah = 0.0145;         //std dev human capital obsolescence shock
39 //sigma = 0.0109;          //std dev TFP innovation epsilon for model B
40
41 //------------------------------------------------------------
42 // 3) Model
43 //------------------------------------------------------------
44 model;
45
46     // TFP shock
47     [name='TFP process']
48     log(z) = rho*log(z(-1)) + eps;
49
```

```
50      // Time constraint
51      [name='Time constraint']
52      n = nbar - l;
53
54      // Human capital growth factor. %to comment out for model B%
55      [name='Human-capital gross growth model A']
56      G = 1 - deltah + (p1*(1 - s) + p2*s)*n;
57
58      // Growth factor for detrending object
59      [name='Detrending factor growth']
60      gx = Gamma^(1/(1 - alpha1)) * G^(alpha2/(1 - alpha1));
61
62      // Stationary production
63      [name='Stationary production']
64      yhat = z * khat(-1)^alpha1 * (s*n)^alpha2;
65
66      // Resource constraint
67      [name='Resource constraint']
68      yhat = chat + ihat;
69
70      // Capital accumulation (stationary)
71      [name='Capital accumulation']
72      khat = ((1 - delta)*khat(-1) + ihat)/gx;
73
74      // Euler 1
75      [name='Euler 1 (capital)']
76      1 = beta * (chat/chat(+1)) * (1/gx) * (1 - delta + alpha1*(yhat(+1)/khat));
77
78      // Euler 2 (s condition)
79      [name='Euler 2 (s)']
80      - alpha2*yhat/(chat*s)
81          = beta * n * (p2 - p1) * (1/G) * (alpha2*yhat(+1)/chat(+1))
82            * (1 - G(+1)/(s(+1)*n(+1)*(p2 - p1)));
83
84      // Euler 3 (n/leisure condition)
85      [name='Euler 3 (n/l)']
86      alpha2*yhat/(chat*n)
87          + beta * (p1*(1 - s) + p2*s) * (1/G) * (
88              (alpha2*yhat(+1)/chat(+1)) * ( 1 - G(+1)/( n(+1)*(p1*(1 - s(+1)) + p2*s(+1)) ) )
89              + (eta/l(+1)) * ( G(+1)/( p1*(1 - s(+1)) + p2*s(+1) ) )
90            )
91          = eta/l;
92      % --------- Model B  2 equations to uncomment-------%
93      %comment out "Human-capital gross growth model A" and uncomment these two:%
94
95      //[name='Human-capital depreciation shock']
96      //log(gammah) = rhoh*log(gammah(-1)) + epsh;
97
98      //[name='Human-capital gross growth model B']
99      //G = 1 - gammah*deltah + (p1*(1 - s) + p2*s)*n;
100
101
102     // --------- Paper-style observables (identities) ----------
103     [name='Labor services (hours): ns = s*n']
104     ns = s*n;
105
106     [name='Labor productivity: q/ns (here q ~ yhat)']
107     yp = yhat/ns;
108
109     [name='Net human capital investment: Dh/h = G-1']
110     dh_over_h = G - 1;
111
112 end;
113
114 //------------------------------------------------------------
115 // 4) Steady state guess (initval)
116 // We choose n and s, then
117 // compute the rest analytically to satisfy several SS identities.
118 //------------------------------------------------------------
119 initval;
120
121     //gammah = 1; %to uncomment for model B%
122
```

```
    z = 1;

    n = 0.90;
    s = 0.90;
    l = nbar - n;

    G  = 1 - deltah + (p1*(1 - s) + p2*s)*n;
    gx = Gamma^(1/(1 - alpha1)) * G^(alpha2/(1 - alpha1));


    khat = (((s*n)^alpha2) / ((gx/beta - 1 + delta)/alpha1))^(1/(1 - alpha1));

    yhat = z * khat^alpha1 * (s*n)^alpha2;

    ihat = gx*khat - (1 - delta)*khat;
    chat = yhat - ihat;

    ns = s*n;
    yp = yhat/ns;
    dh_over_h = G - 1;


end;
resid(1);
steady;
resid;
check;

//---------------------------------------------------------------
// 5) Shocks
//---------------------------------------------------------------
shocks;
    var eps = sigma^2;
    //var epsh = sigmah^2; %to uncomment for model B%
end;

//---------------------------------------------------------------
// 6) Simulation
//---------------------------------------------------------------
stoch_simul(order=1, irf=40, periods=140, drop=100, replic=100);
```